NOCSense — Network Operations Intelligence Platform

Alarm feed × Topology × Correlation ML × NL Assistant × Learning × AI = NOC Intelligence

Powered by NexusAI

South African Telco Operator
1,811 sites · 200,000 alarms/day

PROJECTED ROI

Alarm Noise Elimination, Auto-Remediation,
Operation Optimization and Continuous Learning

90% Daily Opex reduction

~$1.2M Monthly Opex saved

HOW NOCSENSE WORKS

DATA SOURCES

- **Alarm Feed** — 200,000 raw alarms/day
- **Performance Metrics** — Real-time network KPIs
- **Topology Database** — 85,000+ tables
- **Revenue Context** — $2.85M daily across ~2,000 sites

NOCSense AI Engine

Topology-aware correlation + remediation · TMF 642 / 842 compliant

- Standardisation
- Correlation
- Root-Cause ML
- NL Assistant
- Auto-Remediation
- Continuous Learning

INSIGHTS DELIVERED

- **Executive Summary** — NOC KPIs & noise cut
- **Site Rankings** — By revenue impact
- **Root-Cause Analysis** — True failure patterns
- **Action Plans** — Resource + outcome

95.87%	70.2%	4.2 min	Real-time	TMF 642
Alarm noise eliminated at peak	Incidents auto-resolved, no human touch	From alarm to automated fix	Revenue-impact tracking per site	Standardised alarm compliance

THE CHALLENGE

- **Alarm Storm** — Engineers drowning in noise — 8,234 raw alarms/hour at peak, with no prioritisation.
- **Root Causes Hidden** — Just 7 true failure patterns buried inside a 20 million undifferentiated alarm data set.
- **Legacy Data Silos** — Alert feed, performance metrics and an 85,000-table DB — all disconnected, no unified intelligence.
- **No Commercial Awareness** — Faults had no link to revenue impact — the NOC worked by queue order, not by business value.
- **Manual, Reactive Triage** — Every incident sifted by hand from the raw feed — analysts chasing symptoms, not the underlying fault.
- **Slow Resolution** — A multi-hour manual process from alarm to fix, with no learning loop to speed the next one.

THE SOLUTION

A live production pipeline on a South African telco operator's real alarm data — standardising 200,000 alarms/day to TMF 642, correlating them topology-aware, and auto-resolving incidents. No systems replaced.

Auto Data Standardisation
AI agents map 200,000 alarms/day to TMF 642 — 84.1% core compliance, 100% required-field coverage, 98.7% type accuracy.

Topology-Aware Correlation
Temporal + graph + clustering ML group cascading failures — collapsing 8,234 raw alarms/hour into 115 actionable alerts.

Root-Cause Detection
Surfaces the true failure patterns behind the storm — 7 root causes isolated from a 20M-alarm data set.

Natural-Language NOC Assistant
Engineers ask plain-English questions; AI returns site rankings, root causes and action plans — in seconds.

Automated Remediation
Rules + learning engine auto-resolve 70.2% of incidents, with resource assignments and outcome projections.

Continuous Learning
Every human intervention feeds back as training data — auto-resolution improves +5.3% week-on-week.

RESULTS DELIVERED

95.87% Noise Eliminated
8,234 → 115 actionable alerts/hour at peak load.

99% RCA Perfection
Root-cause analysis performed at 99% accuracy across a 20 million-alarm da

70.2% Auto-Resolution Rate
+5.3% week-on-week — the model learns from every engineer action.

4.2 min Avg Resolution Time
For automated incidents — vs. a multi-hour manual process before.

100% TMF Field Coverage
84.1% core compliance and 98.7% data-type accuracy across the feed.

$2.85M Daily Revenue Monitored
Network revenue now tracked in real time, per site.

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